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$$d = \frac{1}{2} g t^2$$

Substituting $g = 9.8$ m $s^{-2}$ we get the depth of the well.



To estimate the error in our calculation we can use another method to measure the depth of the well. Take a long rope and hang the rope inside the well till it touches the bottom. Measure the length of the rope which is the correct depth of the well ($d_{correct}$). Then

$$error = d_{correct} - d$$

$$relative\ error = \frac{d_{correct} - d}{d_{correct}}$$

$$percentage\ of\ relative\ error = \frac{d_{correct} - d}{d_{correct}} \times 100$$

What would be the reason for an error, if any?

Repeat the experiment for different masses and compare the result with $d_{correct}$ every time.

**Note**

If there is water in the well, this method can be used to measure depth of the well till the surface of the water.

**Case (ii): A body thrown vertically upwards**

Consider an object of mass $m$ thrown vertically upwards with an initial velocity $u$. Let us neglect the air friction. In this case we choose the vertical direction as positive y axis as shown in the Figure 2.38, then the acceleration a = –g (neglect air friction) and $g$ points towards the negative $y$ axis. The kinematic equations for this motion are,

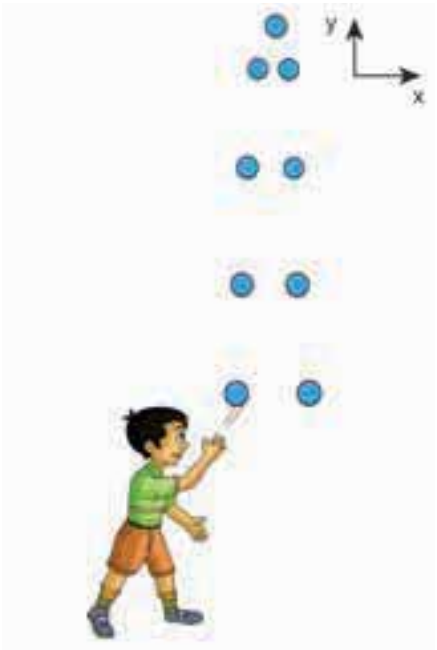



**Figure 2.38.** An object thrown vertically

The velocity and position of the object at any time t are,

$$v = u - gt \qquad (2.20)$$

$$s = ut - \frac{1}{2} g t^2 \qquad (2.21)$$

The velocity of the object at any position y (from the point where the object is thrown) is

$$v^2 = u^2 - 2gy \qquad (2.22)$$